

(CSSE4205) 10 marks

Assignment: Logic Circuit Design and Truth Table Construction (Fall 2025)

Objective

To design and simulate logic circuits for given Boolean expressions using basic logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) and verify results by constructing the corresponding truth tables.

Instructions

1. **Tools Required:**

- Use any logic circuit simulator (Logic simulator)).

2. **For each Boolean expression:**

- a. Construct the **truth table** showing all input combinations and the output.
- b. Draw the **logic circuit diagram** using logic gates.
- c. Simulate the circuit in the chosen simulator to verify output.
- d. Attach screenshots (if done in software) and truth tables.

3. **Submission Format:**

- Title page with name, student ID and section. (University Logo and CCIS)
- Each question should include:
 - Given Boolean Expression
 - Truth Table
 - Logic Diagram (drawn or simulated screenshot)
- File name format: Logic Assignment_<Your ID>.pdf

Name	SET / Questions
	(Total 5 questions for everyone)
Riyam	All 5 sets (Q1)
Rafiaa	All 5 sets (Q3)
Muna	All 5 sets (Q5)
Aisha	All 5 sets (Q2)
Malak	All 5 sets (Q4)
Salim	All 5 sets (Q1)
Khalid	All 5 sets (Q3)
Ahmed	All 5 sets (Q5)
Ali	All 5 sets (Q2)
Osama	All 5 sets (Q4)

Note: Please do not change question no which assigned to you.

SET 1
Q # Boolean Expression 1 $F = (A + B')(A' + C)$ 2 $F = (A \oplus B) + C'$ 3 $F = (A + B)(A + C)'$ 4 $F = A'B + BC' + AB'$ 5 $F = (A + B + C)' + AB$
SET 2
Q # Boolean Expression 1 $F = (A + B')(C + D)$ 2 $F = A'B'C + AB'C + ABC$ 3 $F = (A \oplus B')(C + D)'$ 4 $F = (A + B)(B + C')(A' + C)$ 5 $F = (A' + B')(C' + D)$

SET 3**Q # Boolean Expression**

1 $F = (A + B)(C' + D')$

2 $F = A'B + AB' + CD'$

3 $F = (A + B' + C')(A' + D)$

4 $F = (A \oplus C)(B + D')$

5 $F = (A + B)(C + D)'$

SET 4**Q # Boolean Expression**

1 $F = (A + B')(A' + D)$

2 $F = A'B'C' + AB + AC'$

3 $F = (A \oplus B)(C \oplus D)$

4 $F = A'B + BC + CD'$

5 $F = (A + B' + C')(A' + C + D)$

SET 5**Q # Boolean Expression**

1 $F = (A' + B)(C + D')$

2 $F = (A + B')(C' + D)$

3 $F = (A + C)(B' + D')$

4 $F = A'B'C + AB'C' + ABC$

5 $F = (A + B)'(C + D') + A'C$